7.2.2 Communication and data transmission

Efficient and secure communication and data transmission form the backbone of modern drone operations. From basic remote control in early UAVs to complex real-time data exchange in autonomous systems, communication networks have evolved to support increasingly sophisticated missions. However, as drones become more intelligent and interconnected, maintaining reliable, high-speed, and secure communication links has emerged as one of the most critical challenges in drone evolution.

A primary issue lies in ensuring real-time connectivity between drones, ground stations, and other networked systems, especially in complex or remote environments. Drones operating beyond visual line of sight (BVLOS) or in urban areas often experience signal interference, latency, and bandwidth limitations. These issues can compromise command responsiveness, data integrity, and mission safety. Additionally, terrain obstacles, weather conditions, and electromagnetic interference further degrade signal quality, posing serious challenges for continuous communication (Fang, et al. 2010).

With the increasing demand for high-definition imaging, telemetry, and sensor data streaming, the need for greater bandwidth and lower latency has become essential. The integration of 5G networks, satellite communication, and mesh networking architectures offers promising solutions by enabling seamless, low-latency connections among multiple drones and control systems. These technologies support swarm operations, real-time data analytics, and autonomous decision-making, allowing drones to communicate and coordinate effectively.

However, enhanced connectivity also introduces significant cybersecurity risks. Drones are vulnerable to hacking, jamming, and GPS spoofing, which can lead to unauthorized control or data breaches. To counter these threats,